

Jonathan Z. Schwartz – Data Scientist

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Technical Proficiencies

R (bayesfactor, car, caret, DescTools, dplyr, forecast, ggplot2, lubridate, MASS, R6, RCurl, readr, RPostgreSQL, RSQLite, rvest, shiny, stringr, testthat, tidyverse)
Python (beautifulsoup4, itertools, Matplotlib, NumPy, Pandas, psycpg2, re, scikit-learn, scipy.stats, Seaborn, sqlalchemy, sqlite3, statsmodels, xlswriter)
SDTM (SDISC Certified), SQL, Git, Java, Regular Expressions, Web Scraping, Hypothesis Testing, Probability, Combinatorics

Career Experience

American Thrombosis and Hemostasis Network (ATHN)

Data Scientist

January 2023 – Present

- Contribute to the planning and maintaining of study databases for hemophilia patient data
- Write tools in R and python to automate data science tasks, e.g. to query data from a PostgreSQL database
- Produce data reports and data exports for various stakeholders such as doctors and researchers
- Participant in requirements gathering, ticket grooming, and sprint planning for the data science team
- Transform study data into SDTM format to be submitted to the FDA

Performance Star LLC

Software Engineer – Machine Learning

October 2021 – January 2023

- Contributing to Java and Python code base for a Genetic Programming AI algorithm
- Implemented functionality which tracks summary statistics and displays them to a GUI
- Implemented multi-objective fitness algorithms
- Using Git and Maven for version control and project dependencies.

Wyzant.com

Statistics Tutor

June 2020 – Present

- Freelance tutoring and consulting. Use R daily, helping students and clients learn how to load, clean, and manipulate data, and build many different statistical and machine learning models.
- Gained hundreds of 5-star reviews from students and clients.
- Helped students and working professionals with a wide variety of data analysis, machine learning, and data science assignments and projects in R, python, and SQL.

San Jose State University

San Jose, CA

Statistics Tutor & Grader - Math Department

Spring Semester 2020

- Served as a tutor to teach students in the Math Lab for multiple stats and math classes, including calculus I/II/III, linear algebra, intro to statistics, discrete math, probability theory, applied probability and statistics, R programming, mathematical statistics, design of experiments, linear regression, and stochastic processes.

Statistics Tutor, Student Athlete Success Services

Fall Semester 2019

- Assisted with non-math-major student athletes to provide guidance in introductory statistics subject.

Professional Affiliations

Member of the American Statistical Association (ASA)
Volunteer Member Statistics Without Borders (SWB)
Former Member SJSU Data Science Club
Former Student Member Mathematical Association of America

Publications

- Kopito, G., Schwartz, J., Amblard, J., Filman, R., & Rabern, L. (2023) *MLStar: A System for Synthesis of Machine-Learning Programs*. GECCO '23 Companion: Proceedings of the Companion Conference on Genetic and Evolutionary Computation. 1721 – 1726. <https://dl.acm.org/doi/10.1145/3583133.3596367>
- Ragni, V. M., Callis, J., Daoud, N., Hu, B., Manuel, M., Santos, J., Schwartz, J., Friedman, K. D., Kouides, P., Kuriakose, P., Leavitt, A. D., Lim, M. Y., Machin, N., Recht, M., Chrisentery-Singleton, T. (2023) *Observational Cohort Study of Long-Term Outcomes of Liver Transplantation in Haemophilia*. The Official Journal of the World Federation of Hemophilia, 30(1), 87-97. <https://onlinelibrary.wiley.com/doi/10.1111/hae.14910>

Volunteer Experience

Statistics Without Borders (SWB)

Volunteer

July 2020 – Present

- Contributed to Training Institute Rwanda project by providing PowerPoint slides and recorded lessons to help train Rwandan students in R programming and data science.
- SWB is a volunteer Outreach Group of the American Statistical Association that provides pro bono services in statistics and data science. They work to improve decision making and knowledge in efforts that promote welfare through the proper application of statistical principles and best practices, where access to such resources is limited.

Education & Credentials

Master of Science (M.S.) in Statistics, San Jose State University, San Jose, CA (2018 to 2020), 3.97 GPA

Coursework: Calculus (1, 2, & 3), Discrete Math, Intro to Programming, Applied Probability & Statistics, Linear Algebra, Probability Theory, Regression Theory, R Programming, Stochastic Processes, Mathematical Statistics, Design of Experiments, Time Series Theory & Methods, Categorical Data Analysis, Introduction to Python, Programming & SQL, Statistical Consulting, Computational Statistics.

Bachelor of Arts (B.A.) in Music, Humboldt State University, Arcata, CA (2004 to 2008)

Key Student Projects:

Political Twitter Data Text Mining and Classification Project

Spring 2020

- Deployed NLP Machine Learning algorithms on 80K+ Tweets from members of the U.S. House of Representatives.
- Wrangled & cleaned the data set from Kaggle.com with SQL and mysql.connector python library.
- Utilized scikitlearn to classify Tweets as either Republican or Democrat with 80% Accuracy with TF-IDF model.

Finite Rank Deep Kernel Learning Project sponsored by Intuit

Fall 2019

- Benchmarked & researched new methods in Deep Kernel Learning (DKL).
- Studied background materials & read papers on topics of Gaussian Process Regression, Machine Learning, & Kernel Methods.
- Tested new techniques by using Machine Learning to train kernel functions as well as used GPyTorch (a python package based on PyTorch specifically for Gaussian Process Regression).

Handy Plots: Data Visualization & Analysis R Package

Fall 2018

- Coded & published an R package with various functions to quickly visualize data.
- Submitted to and published on CRAN. Any R user can install my package and use my functions.